OAD: An Unsupervised 3D Masked-Autoencoder for Coastal Ocean Anomaly Detection, Validated Against In-Situ ESP Biology

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Abstract

Harmful algal blooms along the U.S. Pacific Northwest coast are visible in satellite imagery, but predicting downstream biology from single channels like chlorophyll-a has historically worked poorly. We train a 3D convolutional masked-autoencoder on 22 years (2003–2024) of MODIS Aqua imagery over the 41–49° N coastal box. It takes four ocean channels (chlorophyll-a, water clarity, fluorescence, sea-surface temperature) over a 4-day window, sees no bloom labels, and emits one unsupervised "anomaly score" per region per day from its reconstruction error. We evaluate the score on three independent checks. At a 7-day horizon it is the only method we tested with positive R^2 in every region (0.10–0.26), while a matched-complexity PCA baseline sits at zero. The score for the region containing the NEMO mooring tracks in-situ Pseudo-nitzschia cell density measured there during 2016–2018 (Spearman ρ=+0.28, p=0.02; Pearson r=+0.46), validated against an instrument the model never trained on, though the particulate-toxin signal is weaker and not rank-significant. Its summer-mean score for 2014 is the highest on record in three of four regions, matching the documented Pacific marine heatwave. We also document a moderate cloud-cover confound (∼21–24% of score variance in the two highest-amplitude regions) and recommend carrying the in-region valid-pixel fraction as a parallel feature.

1. Introduction

Toxin-producing *Pseudo-nitzschia* blooms happen every year along the U.S. Pacific Northwest (PNW) coast. The toxin they make, domoic acid, builds up in razor clams and shuts down beach harvests [10, 16]. These blooms are not invisible from space, since MODIS Aqua sees the elevated chlorophyll, the warmer-than-average water, the brighter fluorescence, and the changes in water clarity. The problem is that picking out which patches of green water will actually produce toxin from those satellite channels has historically been hard [1,6]. Blooms look different from each other, only some species produce toxin, and PNW cloud cover blocks the sensor a lot of the time.

Single MODIS pixels near the beach are noisy, since clouds, sun glint, river plumes, and shallow-water effects all corrupt the signal. The intuition behind this work is that a per-region *aggregate* should be cleaner than any one pixel, and that a model can learn which spatial and channel structures to aggregate over, rather than just averaging chlorophyll, which gives equal weight to every pixel whether it is informative or not. So instead of training a model to predict a specific target (which needs labels, and bloom labels are expensive), we ask whether an unsupervised model can learn a useful regional summary of "what the ocean looks like" from the satellite record alone. Masked autoencoders [4] are a self-supervised approach where the model has to fill in chunks of the image that have been hidden, and in natural images this forces it to learn higher-level structure than just local texture. We adapt that idea to coastal satellite imagery, where the signal is biophysical (chlorophyll, temperature) rather than categorical (cat vs. dog).

We call the resulting system **OAD** (Ocean Anomaly Detection). It is a 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua over the PNW coast (Fig. 1). The output is a single number per region per day, intended as a starting point for downstream HAB tools. Our contributions:

1. A 3D convolutional autoencoder with three-phase masked-autoencoder training that turns the MODIS cube into one daily score per coastal region.
2. A lead-time comparison showing OAD is the only method tested that is better than predicting the mean at a 7-day horizon in every region, while two natural baselines (matched-complexity PCA and day-of-year climatology) are not.
3. Comparison against in-water Environmental Sample Processor data at the NEMO mooring (2016–2018): the

unsupervised satellite score for the region containing NEMO correlates with measured *Pn* cell density (Spearman ρ=+0.28, p=0.02; Pearson r=+0.46), with a bootstrap CI excluding zero; the toxin signal is weaker and not rank-significant.

4. The summer-mean score for 2014 is the highest of any year in the record in three out of four regions, lining up with the documented Pacific marine heatwave, again without ever being told which year that was.

2. Related Work

Satellite remote sensing for HABs. Operational monitoring has long used satellite chlorophyll-a as a near-real-time bloom indicator [14, 18], but in PNW coastal water the relationship between chl-a and beach-level toxin outcomes is weak [15, 16]. Multi-channel approaches that combine SST, fluorescence, and clarity perform incrementally better [1], but the per-pixel single-day framing remains limiting.

Self-supervised learning for Earth observation. Masked autoencoders [4] train a network to reconstruct large hidden fractions of the input, forcing learning of higher-level structure. In Earth observation, SatMAE [3] and follow-up work [17] apply the principle to optical satellite imagery, primarily for classification. We adapt it to a 3D (time-augmented) coastal ocean cube where the temporal axis lets the model learn dynamics alongside spatial structure, a regime where the dominant signal is biophysical, not land-cover categorical.

In-situ validation via ESP/ChaBa. The Environmental Sample Processor [13] is an autonomous in-water sampler deployed at the NEMO mooring during ChaBa [11], measuring *Pn* cell density (sandwich hybridization) and particulate domoic acid (competitive ELISA) at daily cadence. These datasets are our independent biological validation target. [11] reports general agreement between VIIRS satellite-derived chlorophyll-a and in-situ chl-a at 1 m depth at NEMO, yet finds that elevated *Pn* and domoic acid are not necessarily associated with elevated chlorophyll from offshore source regions, so satellite chl-a tracks biomass but not toxicity. This is part of why we use a multi-channel learned representation rather than a single-channel retrieval.

3. Method

Data. The training cube is built from MODIS Aqua Level-3 daily ocean-color products [12] covering 2003–2024 ($\sim$4,700 daily rolling 8-day composites) on a 321$\times$409 grid at 0.025° resolution. Four channels are used: chl-a, Kd490 (water clarity), normalized fluorescence (nFLH), and SST. A 2D static ocean/land mask excludes land pixels, and NaN pixels (cloud-flagged or otherwise unobserved) are excluded per-pixel from the masked-MSE reconstruction loss.

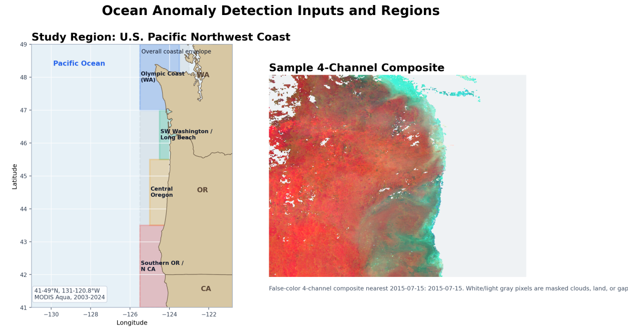

Figure 1. Study region (41–49° N) with example MODIS Aqua 8-day composite. The U.S. PNW coastal box is analyzed at 0.025° resolution on a 321$\times$409 grid. Five alongshore sub-regions are used for downstream analysis; the Olympic Coast region contains the NEMO mooring (47.97° N, 124.97° W) at which ESP in-situ measurements were collected during the 2016–2018 ChaBa deployments.

Architecture. OAD is a 3D ConvAE applied to 64×64 spatial patches tiled over the 321×409 grid, each of shape (C=4, T=4, H=64, W=64) (Fig. 2). The encoder applies three blocks of 3D convolution with stride-2 downsampling, batch normalization, and ReLU activation, reducing the 64×64 field to an 8×8 spatial bottleneck (128$\times$1$\times$8$\times$8) and a latent of dimension 32. The decoder mirrors the encoder with transposed convolutions and upsampling.

Three-phase training. *Phase A* (plain autoencoder) produced reconstruction error that correlated poorly with bloom events, because the model learned to copy textures. *Phase B* added per-region z-score post-processing, which improved scale but not the underlying representation. *Phase C* is masked-autoencoder pretraining [4] with random pixel masking, forcing the model to infer hidden pixels from larger spatial context. Sweeping the mask ratio, we select Phase C with 70% masking (`mae070`) as the headline checkpoint because it is the only variant with positive multi-step forecast skill in all five regions at the 7-day lead (Sec. 4.1). A lower-masking 50% variant (`mae050`) has a modestly cleaner cloud signature (Sec. 4.4) but loses that skill, staying positive in only three of the five regions.

Score derivation. For each daily composite at region R, the trained model produces a per-pixel reconstruction. We average per-pixel reconstruction error over in-region ocean pixels and z-score the result against an in-region historical distribution, producing a single scalar per (region, day). Downstream analyses operate only on this per-region daily scalar, and the convolutional model is not invoked again at consumption time.

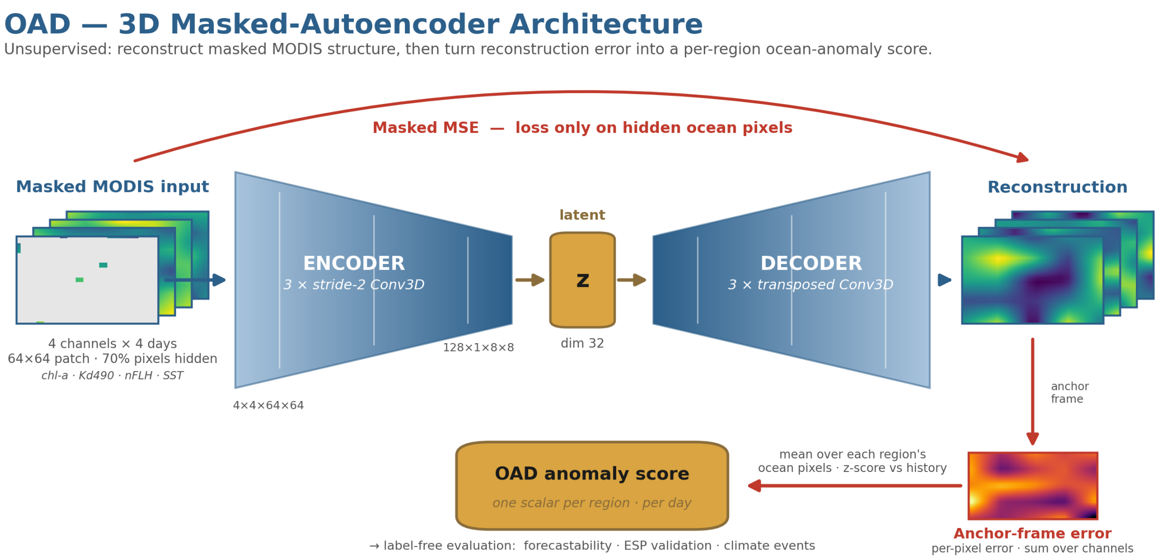

Figure 2. OAD pipeline. Input is a 4-channel (C=4), 4-day-window (T=4), 64×64 patch tiled over the 321×409 grid. The encoder applies three stride-2 3D conv blocks to a latent of dimension 32, and the decoder symmetrically reconstructs the input. During Phase C training, 70% of input pixels are randomly masked and the masked-MSE reconstruction loss is computed only on ocean pixels (cloud-flagged NaNs excluded per-pixel). The anomaly score is the anchor-frame reconstruction error, averaged over each region's ocean pixels and z-scored against history.

4. Results

The OAD score is evaluated along three independent axes: intrinsic lead-time forecastability against baselines (Sec. 4.1), independent in-situ biological validation at the NEMO mooring (Sec. 4.2), and consistency with documented PNW climate events (Sec. 4.3). A short cloud-confound characterization follows (Sec. 4.4).

4.1. Lead-time forecastability vs. baselines

We compare OAD against a matched-dimensionality PCA baseline and a day-of-year climatology baseline, forecasting per-region anomaly state from the past four days on the 2019+ held-out period. Tab. 1 reports results for SW Washington (the highest-amplitude region), and Fig. 3 shows the cross-region picture at the 7-day horizon.

At the honest 7-day lead, OAD is the only method we tested with positive R^2 in every one of the five PNW regions (range 0.10–0.26). PCA, with the same number of dimensions, drops to roughly zero in every region, and the day-of-year climatology baseline actively does worse than predicting the mean in most regions (Fig. 3). An R^2 of 0.10–0.26 is not great in absolute terms, and most of the variance in the next week's ocean state remains unexplained. The point is the gap between OAD and the matched-complexity baselines, not the absolute magnitude.

Does the learning matter, or is the aggregation alone enough? The original motivation for OAD was that single coastal pixels are noisy and a per-region aggregate should be cleaner, so this is worth checking. Fig. 3 answers it directly. The B1 baseline (gray, lightest) is the regional mean of chlorophyll-a over the same pixels OAD sees, z-scored against history, which is the "just average chl-a over the region" strawman. It sits at zero in every region. The B2 baseline (gray, darker) averages all four channels the same way and does *worse*, because the extra channels add noise without a learned weighting. The autoencoder is supplying the "learn what to combine" step that simple regional averaging cannot.

Table 1. Lead-time forecastability R^2, SW Washington region (2019+ held-out). The 1-day result is inflated by 8-day composite overlap (consecutive composites share 7/8 input days), so the honest comparison is at lead $\geq$ 7 days.

Lead	OAD-50	OAD-70	PCA	Climatology
1 day	+0.84	+0.87	−0.13	+0.71
7 days	+0.08	+0.15	−0.13	−0.10
14 days	+0.01	+0.05	−0.13	−0.19

4.2. In-situ validation against ESP/ChaBa

To test whether OAD captures real biology, rather than an idiosyncratic artifact of the unsupervised training objective, we compare it against independent in-situ measurements at the same time and place. We use ESP-derived *Pn* cell

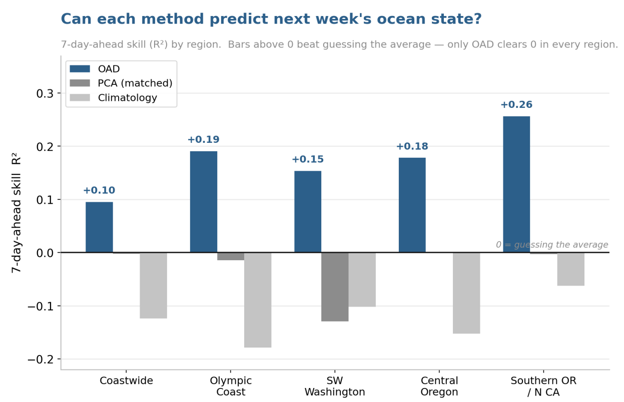

Figure 3. **Held-out 7-day-ahead forecast skill (R^2), by region (2019+).** Each method predicts the next week's regional anomaly state from the past 4 days, and bars above zero do better than just guessing the mean. OAD (purple, the 70%-mask variant) is positive in all five regions (0.10–0.26), which is not a huge number, but every other method we tried sits at or below zero. PCA (gray) is the right control because it has the same number of dimensions as the autoencoder, so the gap is not just "more parameters help." Day-of-year climatology (light gray) is worse than the mean. The lower-masking OAD-50 variant is positive in only 3 of 5 regions, since it trades multi-step skill for a slightly cleaner cloud-cover signature (Sec. 4.4), which is why we report and ship mae070 rather than mae050.

density and particulate domoic acid (pDA) from the 2016–2018 ChaBa deployments at the NEMO mooring (47.97° N, 124.97° W) [11]. NEMO lies within our Olympic Coast region, and both quantities are sampled at that single location, so we compare them against that one region's OAD score. Because cell-density and toxin distributions are heavy-tailed (a few bloom days dominate), we report both Pearson r on raw values *and* the rank-based Spearman ρ, each with 95% non-parametric bootstrap CIs (2,000 resamples, seed = 42), and they appear in Tab. 2.

The *Pn*-cell signal is the robust one. The OAD score for the NEMO region tracks measured cell density with Pearson $r = 0.46$ ($p = 3\times10^{-5}$) and, more conservatively, Spearman $\rho = 0.28$ ($p = 0.016$, bootstrap CI excluding zero). Fig. 4 shows this directly, as median cell density rises monotonically across terciles of the OAD score ($5.1 \rightarrow 6.4 \rightarrow 7.6 \times 10^4$ cells L^{-1}). The particulate-toxin signal is weaker. The raw Pearson $r = 0.34$ is carried by a handful of high-toxin days, and the rank correlation is *not* significant (Spearman $\rho = 0.19$, $p = 0.07$, CI spanning zero). We therefore treat *Pn* cell density as the validated target and pDA as a suggestive but unconfirmed secondary. The cell-density correlation is moderate, and there is plenty of bloom-day-to-bloom-day biology the satellite misses, but the model never saw these in-water measurements, so a signal that survives a rank test is informative.

Table 2. OAD score for the Olympic Coast region (which contains NEMO) vs. the in-situ ESP measurements taken at NEMO, 2016–2018 (*Pn* cells by SHA assay; pDA by cELISA). Pearson is on raw values; Spearman is the rank correlation, robust to the heavy-tailed bloom-spike days. Brackets are 95% bootstrap CIs (2,000 resamples, seed = 42). The cell-density signal survives the rank test; the toxin signal does not.

Measurement	**Pearson r**	**Spearman ρ**	N
Pn cells	**+0.46** [.17,.66]	**+0.28** [.01,.51]	76
pDA (toxin)	+0.34 [.02,.54]	+0.19 [−.03,.39]	90

The time-series overlay (Fig. 6) shows OAD and ESP peaks tending to move together across the five 2016–2018 deployment windows. The 2017 spring deployment is the clearest case, where OAD ramps up several days before the ESP-measured *Pn* peak, which suggests it may be useful as a leading indicator for when to deploy in-water samplers.

4.3. Agreement with documented PNW climatology

The unsupervised score also recovers well-established PNW coastal ocean patterns without ever having been told about specific events. Three patterns emerge.

Seasonal cycle. The score rises sharply in late spring (April–May), peaks in mid-summer, and declines through autumn across all five regions, matching canonical PNW *Pn* bloom phenology [8, 15, 16]. Amplitude is largest in SW Washington (Columbia River plume $\times$ shelf upwelling) and Olympic Coast (Juan de Fuca eddy retains nutrient-rich water [5]).

The 2014 marine heatwave stands out. One test of the score's climate sensitivity is whether it picks out the year of the 2014–16 Pacific marine heatwave ("the Blob" [2]), which triggered the largest *P. australis* bloom on record [9, 10]. Fig. 5 plots the monthly-mean OAD score per region across 2003–2024. In Olympic Coast, Central Oregon, and Southern OR/N. California, 2014 is the highest summer-mean of all 19 measured years, and in SW Washington it is the second-highest. The model was never told which years had blooms, so this ranking comes entirely from the satellite imagery itself.

Cross-region coherence with heterogeneity. Regional scores correlate at > 0.5 on monthly means, consistent with PDO-modulated basin-scale upwelling and ENSO teleconnections [7], but each region preserves its own high-frequency signal from local eddies, plumes, and wind events. This is the kind of structure we would expect from a representation that reflects PNW coastal dynamics.

4.4. Cloud-cover confound

Per-region Pearson correlation between the OAD score and the in-region valid-pixel fraction is moderate. For our headline mae070 checkpoint, SW Washington shows $r \approx$

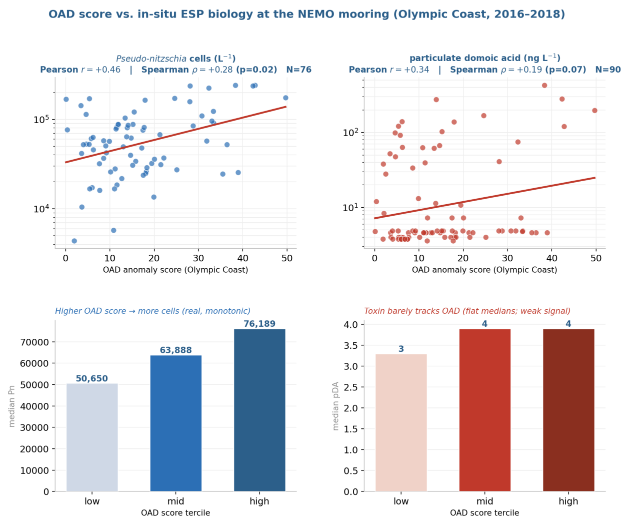

Figure 4. **OAD score for the Olympic Coast region vs. the in-situ biology measured at the NEMO mooring it contains (2016–2018). Top:** per-day scatter (one dot per matched day; red line is a log-linear fit for visualization; y log-scaled for readability) annotated with both Pearson r and the rank-based Spearman ρ. **Bottom:** median ESP value within terciles of the OAD score. Cell density (left), the validated signal, rises monotonically with the score, while the toxin (right, pDA) is essentially flat across terciles, consistent with its non-significant rank correlation.

$+0.49$ ($\sim$24% of variance covaries with cloud cover) and Olympic Coast shows $r \approx +0.46$ ($\sim$21%). The cleaner-cloud mae050 variant is 4–9 percentage points lower, but we do not select it because it loses multi-step forecast skill (Sec. 4.1). Downstream applications should carry the in-region valid-pixel fraction as a parallel feature so consuming models can learn to discount cloudy weeks.

5. Discussion

The three checks each show something modest but real. The forecastability R^2 is small in absolute terms (0.10–0.26), but every baseline we tried at the same complexity drops to zero or below, so the gap, not the absolute number, is the result. The ESP cell-density correlation (Pearson 0.46, Spearman 0.28) is moderate but survives a rank test, while the toxin correlation does not, and the model was never trained on those measurements, so even a modest consistent signal is informative. The 2014 heatwave-year ranking comes out cleanly without any temporal labels. None of this proves the score is a finished bloom-prediction tool. It does show that an unsupervised representation of the satellite cube is picking up structure that linear baselines miss and that lines up with real biology at the source.

What this paper does *not* show is whether the source-region OAD score predicts beach-level shellfish toxicity. That depends on a chain of additional physics, including wind direction, onshore transport, which *Pn* species is present and whether it is producing toxin, and how much each clam happens to filter. We test only the offshore signal here.

Limitations. (1) the 2009–2011 MODIS coverage gap leaves wide uncertainty in some regions, (2) the cloud-confound is partial ($\sim$21–24% variance), (3) in-situ validation uses only one mooring (NEMO) with 76–90 daily samples, and (4) regions are hand-defined, so learning regions via clustering on the latent is a natural extension.

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A. Supplementary Figures

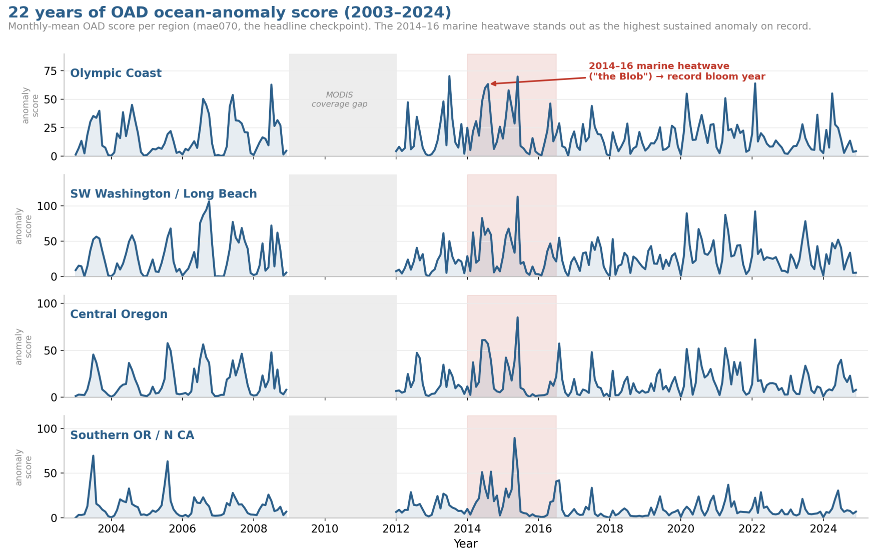

Figure 5. **Monthly-mean OAD score per region across the full 2003–2024 record.** The 2014–16 Pacific marine heatwave ("the Blob," shaded) stands out as the highest sustained anomaly in the series. Quantitatively, 2014 is the highest summer (May–September) mean of any year on record in three of the four regions, and the second-highest in the fourth, without the model ever being shown a bloom-year label. The gray band marks the 2009–11 MODIS Aqua coverage gap, where there is no data and the line is broken across it.

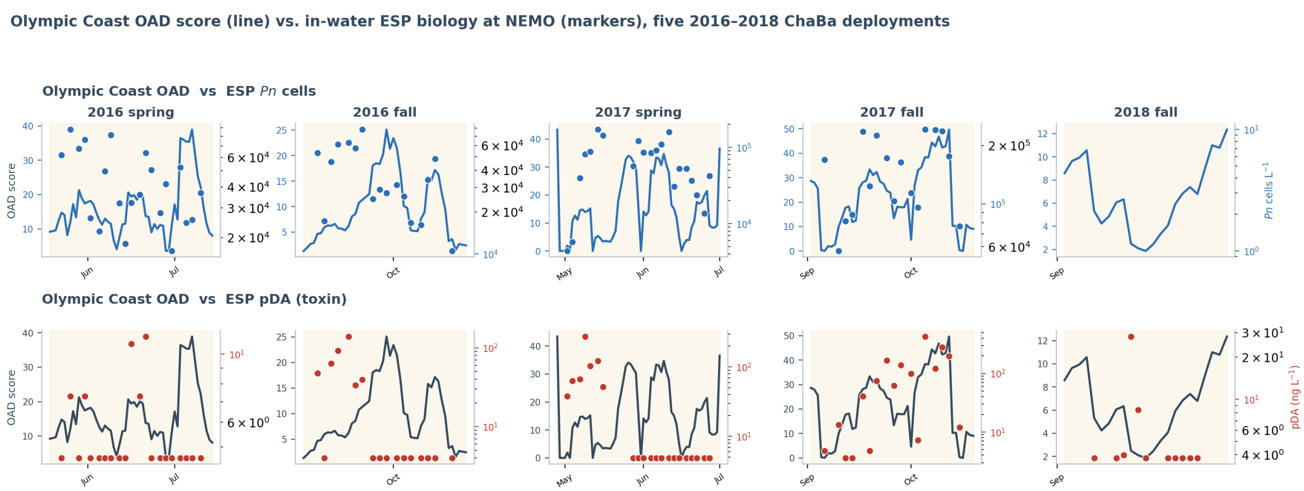

Figure 6. **Day-by-day overlay of the Olympic Coast OAD score (line, left axis) and the in-water ESP measurements at NEMO (markers, right axis, log scale) during the five 2016–2018 ChaBa deployments.** Yellow bands mark each deployment window. The top row is ESP *Pn* cells per liter, and the bottom row is ESP particulate domoic acid in ng/L. When the satellite score goes up, the in-water biology tends to follow. The 2017 spring panel (third column) is the cleanest example, where OAD starts rising several days before the ESP measures the *Pn* peak, which is the kind of behavior you would want from an early-warning signal.